Essential in Civil 3D (40 Hours)

S.NO	Session	Topics	Type	Duration (Hours)
1	Session 1	**Introduction** Land Survey Land Development Transportation Design AutoCAD Civil 3D **Working Environment** Work Space Tool Space Object Object Style Label Style Creating Template Creating Reports	Theory/Demo	2.0 Hrs
2	Session 2	Exercises included for Practice	Practical	2.0 Hrs
3	Session 3	**Working with Point Data** Creating and Editing Points Miscellaneous Intersection Slope Interpolate Point Styles Point Group Import/Export Points	Theory/Demo	2.0 Hrs
4	Session 4	Exercises included for Practice	Practical	2.0 Hrs
5	Session 5	**Surface** How to create a surface based on Points Modifying Surface Surface Styles Surface Label Styles Create Points from Surface	Theory/Demo	2.0 Hrs
6	Session 6	Exercises included for Practice	Practical	2.0 Hrs
7	Session 7	**Survey: Total Station** Creating Survey Database Creating Survey Network Creating Figure Styles and Prefix Importing Field Book Working with Survey Data Creating Surface **LiDAR Survey** Introduction to LiDAR How to Import LiDAR Survey Data Creating Surface **Working with DEM files**	Theory/Demo	2.0 Hrs
8	Session 8	Exercises included for Practice	Practical	2.0 Hrs
9	Session 9	**Surface Analysis** Direction Elevation Slope Slope Arrow Water Shed Water Drop Catchment Area Legend Table	Theory/Demo	2.0 Hrs
10	Session 10	Exercises included for Practice	Practical	2.0 Hrs
11	Session 11	**Earthwork Calculation** Create surface by Breaklines Contours Drawing Objects Alignment Creation Horizontal Profile View generation Customising Profile Views Create a Volume Surface Volume Calculation	Theory/Demo	2.0 Hrs

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		Grading		
		Xref		
		Combining Surface from multiple files		
		Global Coordinates		
12	Session 12	Exercises included for Practice	Practical	2.0 Hrs
13	Session 13	**Parcel**	Theory/Demo	2.0 Hrs
		Site		
		Creating Parcels		
		From Object		
		By Layout		
		Creating Right Of Way		
		Editing Parcels		
		Labelling Parcels		
		Creating Tables		
14	Session 14	Exercises included for Practice	Practical	2.0 Hrs
15	Session 15	**Alignment Design as per IRC**	Theory/Demo	2.0 Hrs
		Design Criteria Editor		
		Alignment from Polyline		
		Alignment Style		
		Alignment Label Style		
		Design Criteria		
		Design Check Set		
		Alignment by Layout		
		Edit Alignment Geometry		
		Tables		
		Best Fit Alignment		
		Create Offset Alignment		
		Create Widening		
16	Session 16	Exercises included for Practice	Practical	2.0 Hrs
17	Session 17	**Corridor and Corridor Surface**	Theory/Demo	2.0 Hrs
		Create Corridor		
		View/Edit Corridor Sections		
		Super Elevation		
		Corridor Surface		
		Corridor Surface Boundaries		
18	Session 18	Exercises included for Practice	Practical	2.0 Hrs
19	Session 19	**Junction Design**	Theory/Demo	2.0 Hrs
		Manual Junction Design (based on versions before 2010)		
		Automatic Junction Design		
		Customise the Assemblies		
		Corridor Surface		
		Quantity Takeoff		
		Sample Lines		
		Section Views		
		Compute Materials		
		Earthwork		
		Material Quantity Takeoff		
		Creating New Criteria		
20	Session 20	Exercises included for Practice	Practical	2.0 Hrs